

DATA VORTEX 2026

Round 1 — Phase 2

Selected SQL Queries

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E3 — Easy: Average Engagement by Platform

```
SELECT
  platform,
  ROUND(AVG(likes), 2) AS avg_likes,
  ROUND(AVG(shares), 2) AS avg_shares,
  ROUND(AVG(comments), 2) AS avg_comments,
  ROUND(AVG(likes + shares + comments), 2) AS avg_total_engagement
FROM posts
WHERE platform IS NOT NULL
  AND platform <> 'Unknown'
GROUP BY platform
ORDER BY avg_total_engagement DESC;
```

M3 — Medium: Most Active Users

```
SELECT
  u.user_id,
  u.follower_count,
  u.location,
  COUNT(p.post_id) AS post_count,
  SUM(p.likes + p.shares + p.comments) AS total_engagement
FROM users u
JOIN posts p ON u.user_id = p.user_id
GROUP BY u.user_id, u.follower_count, u.location
ORDER BY post_count DESC, total_engagement DESC
LIMIT 10;
```

H4 — Hard: Follower-to-Engagement Anomaly

```
WITH user_totals AS (
  SELECT
    u.user_id,
    u.location,
    u.follower_count,
    SUM(p.likes + p.shares + p.comments) AS total_engagement
  FROM users u
  JOIN posts p ON u.user_id = p.user_id
  GROUP BY u.user_id, u.location, u.follower_count
```

```
),  
ranked AS (  
  SELECT *,  
    NTILE(10) OVER (ORDER BY total_engagement DESC) AS engagement_decile  
  FROM user_totals  
)  
SELECT  
  user_id,  
  location,  
  follower_count,  
  total_engagement,  
  engagement_decile  
FROM ranked  
WHERE follower_count < 5000  
  AND engagement_decile = 1  
ORDER BY total_engagement DESC;
```